

8086 Memory Addressing Modes

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Lecture References:

▶ **Book:**

- ▶ *Microprocessors and Interfacing: Programming and Hardware*, Chapter # 2, **Author:** Douglas V. Hall
- ▶ *Assembly Language Programming And Organization of the IBM PC*, Chapter # 4, **Author:** Ytha Yu, Charles Marut.

▶ **Lecture Materials:**

- ▶ *IBM PC Organization*, CAP/IT22 I
- ▶ M.A. Sattar, Microprocessor and Microcontroller

Memory Model Segmentation

▶ **.model small**

- ▶ Most widely used memory model.
- ▶ The code must fit in 64k.
- ▶ The data must fit in 64k.

▶ **.model medium**

- ▶ The code can exceed 64k.
- ▶ The data must fit in 64k.

▶ **.model compact**

- ▶ The code must fit in 64k.
- ▶ The data can exceed 64k.

▶ **.medium and .compact are opposites.**

Basic Structure of Assembly Language Program

```
.MODEL SMALL/COMPACT/MEDIUM
.STACK 100H
.DATA
----
----
.CODE
MAIN PROC
    MOV AX, @DATA
    MOV DS, AX
....
---
MAIN ENDP
    END MAIN
RET
```

How to Define Data Segment

- ▶ **db:** define byte
- ▶ **dw:** define word (2 bytes)
- ▶ **equ:** define a numeric constant to a name

.data

data_byte DB 255 ; Define a data with 1-byte

data_word DW 512 ; Define a data with 2-bytes

data_string DB "Hello"; define 5 consecutive bytes with ASCII values

max_int equ 65535 ; define maximum value of integer =65535

How to Define Stack Segment

.STACK size

Where, size is an optional number that specifies the stack area size in bytes.

.STACK 100h

Here, it sets 00100h bytes for the stack. If size is omitted then 1 KB size is set aside for the stack.

Addressing Mode and It's Categories

- ▶ The different ways in which a microprocessor can access data are referred to as its addressing modes.
- ▶ Addressing modes of 8086 Microprocessor are categorized as:
 - ▶ *Addressing Data*
 - ▶ *Addressing Program codes in memory*
 - ▶ *Addressing Stack in memory*
 - ▶ *Addressing I/O*
 - ▶ *Implied addressing*

1. Addressing Data

- I. Immediate addressing
- II. Direct addressing
- III. Register [direct] addressing
- IV. Register indirect addressing
- V. Base-plus-index addressing
- VI. Register relative addressing
- VII. Base-relative-plus-index addressing

1. Addressing Data

i. Immediate addressing

- ▶ Data is immediately given in the instruction

MOV BL, 44
MOV BX, 256

ii. Register [Direct] addressing

- ▶ Data is in a register (here BX/BL register contains the data)

MOV AX, BX
MOV AL, BL

1. Addressing Data

iii. Register [Indirect] addressing

- ▶ Register supplies the address of the required data

MOV AX, [BX] ;

Store the value of (DSx10h+BX) in AX

iv. Direct addressing

- ▶ Data address is directly given in the instruction

MOV AX, [1234H] ; AX $\leftarrow$ Value in (DSx10h+1234)

MOV AX, DATA_VAR ; DATA_VAR is a 16-bit variable

1. Addressing Data

v. **Base-plus-index addressing**

- ▶ Base register is either BX or BP
- ▶ Index register is either DI or SI

MOV DX, [BX+DI] ;

Store the value of (DSx10h+BX+DI) in DX

vi. **Register relative addressing**

- ▶ Register can be a base (BX, BP) or an index register (DI,SI)
- ▶ Mainly suitable to address array data

MOV DX, [BX+2] ;

Store the value of (DSx10h+BX+2) in DX

1. Addressing Data

vii. **Base-relative-plus-index addressing**

- ▶ Suitable for array addressing

MOV DX, [BX+DI+10] ;

Store the value of $(DS \times 10h + BX + DI + 10)$ in DX

2. Addressing Program Codes in Memory

- ▶ Used with JMP and CALL instructions
- ▶ 3 distinct forms:
 - ▶ Direct
 - ▶ Indirect
 - ▶ Relative

2. Addressing Program Codes in Memory

- ▶ Address is directly given in the instruction

JMP 1000:0000

or **JMP doagain** ; doagain is a **label** in code

CALL 1000:0000

or **CALL doagain** ; doagain is a **procedure** in code

- ▶ Often known as *far* jump or *far* call

2. Addressing Program Codes in Memory

- ▶ Address can be obtained from
 - ▶ **a)** any GP registers (AX,BX,CX,DX)
 - ▶ **b)** any relative registers ([BP],[BX],[DI],[SI])
 - ▶ **c)** any relative register with displacement

JMP [DI] ; Jump to memory location (CSx10h + DI)

CALL [BX] ; Call the content in memory location
(CSx10h + BX)

3. Addressing Stack in Memory

- **PUSH** and **POP** instructions are used to move data to and from stack (in particular from stack segment).

PUSH AX

PUSH BX

POP AX

POP BX

- (Alternative for SWAP Operation: ***XCHG AX, BX***)
- **CALL** also uses the stack to hold the return address for procedure.

4. Addressing Input and Output Port

- ▶ IN and OUT instructions are used to address I/O ports
- ▶ Could be ***direct addressing***

IN AL, 05h ; Here 05h is a input port number

- ▶ or ***indirect addressing***

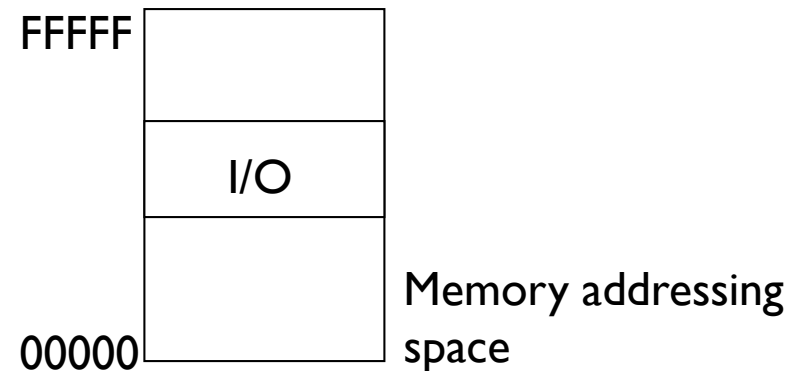
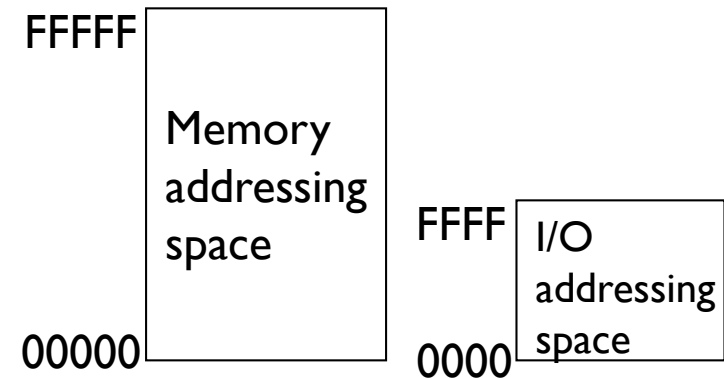
OUT DL, AL ; DL contains the address of I/O port

- ▶ Only DL register can be used to point a I/O port

4. Addressing Input and Output Port

► Mapped I/O

- A device is mapped to a memory location. Sending data to the particular location causes interaction with the device.
- 16-bit I/O port addresses
- Uses same memory read and write signals
- Most processors use this I/O mapping



Memory-mapped I/O

5. Implied Addressing

- ▶ No explicit address is given with the instruction
- ▶ implied within the instruction itself
- ▶ Examples:

CLC ; clear carry flag

HLT ; halts the program

RET ; return to DOS is equivalent of

MOV AH, 4Ch

INT 21h

Thank You !!

